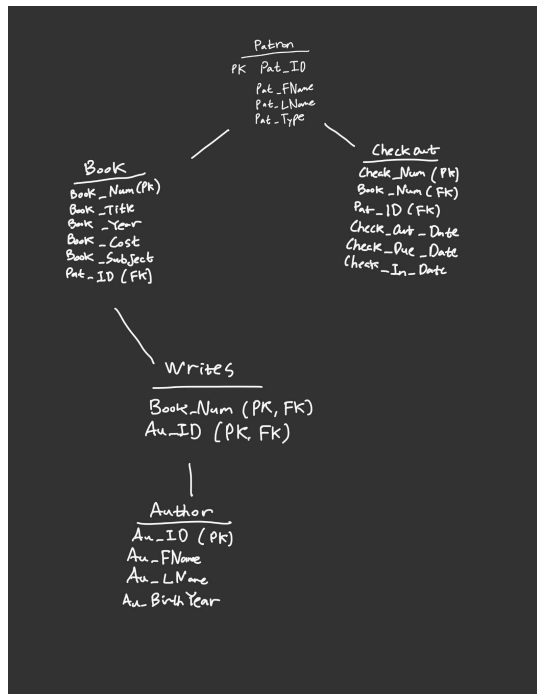


ERD



#56

BOOK_TITLE ▾	BOOK_COST ▾	BOOK_YEAR ▾
Beginner's Guide	59.95	2014
Beyond the Data	69.95	2016
C# in Middlewa	59.95	2015
Capture the Cl	69.95	2016
Cloud-based M	69.95	2015
Coding Style fo	49.95	2017
Conceptual Pro	59.95	2015
Database in the	79.95	2014
DATABASES in	129.95	2015
iOS Programmi	79.95	2015
J++ in Mobile A	49.95	2015
JAVA First Step	49.95	2015
Mastering the	89.95	2015
Reengineering	89.95	2016
Shining Throug	109.95	2016
Starlight Applic	69.95	2016
The Golden Ro	119.95	2016
Thoughts on Re	59.95	2016
Virtual Program	79.95	2016
What You Alwa	49.95	2016
*	0	0

```

SELECT BOOK_TITLE, BOOK_COST, BOOK_YEAR
FROM BOOK
ORDER BY BOOK_TITLE;

```

#60

```
SELECT DISTINCT BOOK_YEAR
FROM BOOK
ORDER BY BOOK_YEAR;
```

BOOK_YEAR ▾
2014
2015
2016
2017

#65

```
SELECT BOOK_NUM, BOOK_TITLE, BOOK_COST
FROM BOOK
WHERE BOOK_COST = 59.95
ORDER BY BOOK_NUM;
```

BOOK_NUM ▾	BOOK_TITLE ▾	BOOK_COST ▾
5235	Beginner's Guide	59.95
5238	Conceptual Pro	59.95
5242	C# in Middlewa	59.95
5251	Thoughts on Re	59.95
* 0		0

#70

```
SELECT AU_ID, AU_FNAME, AU_LNAME, AU_BIRTHYEAR
FROM AUTHOR
WHERE AU_BIRTHYEAR BETWEEN 1980 AND 1989
ORDER BY AU_ID;
```

AU_ID ▾	AU_FNAME ▾	AU_LNAME ▾	AU_BIRTHYE ▾
218	Rachel	Beatney	1983
383	Neal	Walsh	1980
394	Robert	Lake	1982
438	Perry	Pearson	1986
460	Connie	Paulsen	1983
581	Manish	Aggerwal	1984
603	Julia	Palca	1988
* 0			0

#75

```
SELECT AU_ID, AU_FNAME, AU_LNAME
FROM AUTHOR
WHERE AU_BIRTHYEAR IS NOT NULL
ORDER BY AU_ID;
```

AU_ID ▾	AU_FNAME ▾	AU_LNAME ▾
185	Benson	Reeves
218	Rachel	Beatney
251	Hugo	Bruer
273	Reba	Durante
284	Trina	Tankersly
383	Neal	Walsh
394	Robert	Lake
438	Perry	Pearson
460	Connie	Paulsen
581	Manish	Aggerwal
592	Lawrence	Sheel
603	Julia	Palca
* 0		

#80

```
SELECT Count(*) AS [Available Books]
FROM BOOK
WHERE BOOK_NUM NOT IN
(SELECT BOOK_NUM
FROM CHECKOUT
WHERE CHECK_IN_DATE IS NULL);
```

Available Bc ▾
14

#81

SELECT Max(BOOK_COST) AS [Most Expensive] FROM BOOK;	Most Expensive ▾ 129.95
---	----------------------------

#83

SELECT Count(PAT_ID) AS [Different Patrons] FROM (SELECT DISTINCT PAT_ID FROM CHECKOUT);	Different Patrons ▾ 33
---	---------------------------

#89

SELECT BOOK_NUM, BOOK_TITLE & " (" & BOOK_YEAR & ")" AS BOOK, BOOK_SUBJECT FROM BOOK ORDER BY BOOK_NUM;	BOOK_NUM ▾	BOOK ▾	BOOK_SUBJECT ▾		
	5235	Beginner's Guide to Programming			
	5236	Database in the Cloud			
	5237	Mastering the Oracle Database			
	5238	Conceptual Programming			
	5239	J++ in Mobile Applications Programming			
	5240	iOS Programming			
	5241	JAVA First Steps Programming			
	5242	C# in the Middle: Middleware			
	5243	DATABASES in the Cloud			
	5244	Cloud-based Middleware			
	5245	The Golden Rules of Middleware			
	5246	Capture the Cloud			
	5247	Shining Through Programming			
	5248	What You Always Wanted from a Database			
	5249	Starlight Applications in the Cloud			
	5250	Reengineering Middleware			
	5251	Thoughts on Re-engineering Programming			
	5252	Beyond the Database			
	5253	Virtual Programming			
	5254	Coding Style for Programming			
	*	0			

#90

SELECT AUTHOR.AU_LNAME, AUTHOR.AU_FNAME, WRITES.BOOK_NUM FROM (AUTHOR INNER JOIN WRITES ON AUTHOR.AU_ID = WRITES.AU_ID) INNER JOIN BOOK ON WRITES.BOOK_NUM = BOOK.BOOK_NUM ORDER BY AUTHOR.AU_LNAME, AUTHOR.AU_FNAME, WRITES.BOOK_NUM;	AU_LNAME ▾	AU_FNAME ▾	BOOK_NUM ▾	
	Aggerwal	Manish	5242	
	Beatney	Rachel	5240	
	Bruer	Hugo	5243	
	Bruer	Hugo	5246	
	Chiang	Xia	5244	
	Chiang	Xia	5249	
	Chiang	Xia	5252	
	Durante	Reba	5235	
	Lake	Robert	5245	
	Lake	Robert	5247	
	McGill	Rachel	5241	
	McGill	Rachel	5254	
	Palca	Julia	5238	
	Paulsen	Connie	5239	
	Paulsen	Connie	5241	
	Paulsen	Connie	5251	
	Pearson	Perry	5250	
	Reeves	Benson	5237	
	Reeves	Benson	5253	
	Salvadore	Carmine	5239	
	Salvadore	Carmine	5248	
	Sheel	Lawrence	5239	
	Tankersly	Trina	5244	
	Walsh	Neal	5236	
	Walsh	Neal	5250	
	*			

#98

```
SELECT
    BOOK.BOOK_NUM,
    BOOK.BOOK_TITLE,
    Count(CHECKOUT.CHECK_NUM) AS [Times Checked Out]
FROM BOOK
LEFT JOIN CHECKOUT
    ON BOOK.BOOK_NUM = CHECKOUT.BOOK_NUM
GROUP BY
    BOOK.BOOK_NUM,
    BOOK.BOOK_TITLE
ORDER BY
    Count(CHECKOUT.CHECK_NUM) DESC,
    BOOK.BOOK_TITLE;
```

BOOK_NUM	BOOK_TITLE	Times Check
5236	Database in the Cloud	12
5235	Beginner's Guide to Programming	9
5240	iOS Programming	7
5238	Conceptual Programming	6
5237	Mastering the Database	5
5252	Beyond the Database	4
5242	C# in Middleware	4
5246	Capture the Cloud	4
5244	Cloud-based Middleware	4
5254	Coding Style for Programming	4
5249	Starlight Application	4
5248	What You Always Wanted	3
5243	DATABASES in the Database	2
5239	J++ in Mobile Application	0
5241	JAVA First Step Programming	0
5250	Reengineering Middleware	0
5247	Shining Through Programming	0
5245	The Golden Rule of Middleware	0
5251	Thoughts on Reengineering	0
5253	Virtual Programming	0

#106

```
SELECT AU_ID, AU_FNAME, AU_LNAME
FROM AUTHOR
WHERE AU_ID NOT IN
    (SELECT WRITES.AU_ID
    FROM (WRITES
    INNER JOIN BOOK ON WRITES.BOOK_NUM = BOOK.BOOK_NUM)
    WHERE BOOK.BOOK_SUBJECT = 'Programming')
ORDER BY AU_LNAME;
```

AU_ID	AU_FNAME	AU_LNAME
581	Manish	Aggerwal
251	Hugo	Bruer
262	Xia	Chiang
438	Perry	Pearson
284	Trina	Tankersly
383	Neal	Walsh
0		

#107

```
SELECT
    B1.BOOK_NUM,
    B1.BOOK_TITLE,
    B1.BOOK_SUBJECT,
    (SELECT Avg(B2.BOOK_COST)
    FROM BOOK AS B2
    WHERE B2.BOOK_SUBJECT = B1.BOOK_SUBJECT) AS AVG COST,
    B1.BOOK_COST -
    (SELECT Avg(B2.BOOK_COST)
    FROM BOOK AS B2
    WHERE B2.BOOK_SUBJECT = B1.BOOK_SUBJECT) AS DIFFERENCE
FROM BOOK AS B1
ORDER BY B1.BOOK_TITLE;
```

BOOK_NUM	BOOK_TITLE	BOOK_SUBJECT	AVG COST	DIFFERENCE
5235	Beginner's Guide to Programming	Programming	66.61666666666666	-6.666666666666666
5252	Beyond the Database	Database	84.95	-15
5242	C# in Middleware	Middleware	89.95	-30
5246	Capture the Cloud	Cloud	72.45	-2.5
5244	Cloud-based Middleware	Cloud	72.45	-2.5
5254	Coding Style for Programming	Programming	66.61666666666666	#####
5238	Conceptual Programming	Programming	66.61666666666666	-6.666666666666666
5236	Database in the Cloud	Cloud	72.45	7.5
5243	DATABASES in the Database	Database	84.95	45
5240	iOS Programming	Programming	66.61666666666666	13.333333333333334
5239	J++ in Mobile Application	Programming	66.61666666666666	#####
5241	JAVA First Step Programming	Programming	66.61666666666666	#####
5237	Mastering the Database	Database	84.95	5
5250	Reengineering Middleware	Middleware	89.95	0
5247	Shining Through Programming	Programming	66.61666666666666	43.333333333333334
5249	Starlight Application	Cloud	72.45	-2.5
5245	The Golden Rule of Middleware	Middleware	89.95	30
5251	Thoughts on Reengineering	Programming	66.61666666666666	-6.666666666666666
5253	Virtual Programming	Programming	66.61666666666666	13.333333333333334
5248	What You Always Wanted	Database	84.95	-35